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MODULE 9
STATISTICAL TESTS FOR MEANS

"Better Analysts Make a Smarter Tomorrow..."

Learn Apply Grow

"Statistics turns data into opportunity."

STATISTICAL TESTS FOR MEANS

From Concepts to Real-World Business Decisions

- Simple Explanations
- Real-World Examples
- Excel Calculations
- Interview Preparation

What is this Module About?

This module focuses on statistical tests used to compare means. You'll learn when to use Z-tests and t-tests, how to perform one-sample, two-sample (independent and paired) tests, check assumptions, implement calculations in Excel, and interpret results in real business scenarios.

> "It's not just about numbers, it's about making better business decisions with data."

Why is it Important?

- Helps us validate business claims.
- Used to compare averages (e.g., sales, conversion rate, delivery time, ratings).
- Guides decision-making before and after changes.
- Widely used in business, research, analytics, and operations.
- Supports A/B testing, process improvements, and policy analysis.

Real-World Impact

- Improve website and marketing performance.
- Measure the impact of campaigns and promotions.
- Evaluate process and product improvements.
- Compare performance between groups or time periods.
- Make informed financial and operational decisions.
- Analyze healthcare and clinical trial results.

Visual Overview: Comparing Means

Z-Test

- Used when population standard deviation (σ) is known.
- Commonly used for large samples (n ≥ 30).
- Based on standard normal distribution (Z).

t-Test

- Used when population standard deviation (σ) is unknown.
- Commonly used for small samples (n < 30).
- Based on t-distribution (with degrees of freedom).

Compare Two Means

- Independent samples (two different groups).
- Paired samples (same group before vs after).
- Use Z-test or t-test based on data and assumptions.

Topics We Will Cover (12 Topics)

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Business Use Cases at a Glance

E-commerce	Marketing	Manufacturing	HR Analytics	Finance	Healthcare
Is the new website design increasing conversion rate?	Did the new campaign improve average sales?	Has the average defect rate reduced after a process improvement?	Is the average employee satisfaction higher after a new policy?	Has the average transaction value increased after a new pricing strategy?	Is the new treatment more effective than the existing one?

1 Z-Test: When and Why?

Testing a Population Mean When Standard Deviation is Known

What is a Z-Test?

A Z-test is a statistical test used to check whether the sample mean is significantly different from a hypothesized population mean when the population standard deviation (σ) is known.

> "Z-test helps us decide whether an observed difference in the sample mean is real or just due to random chance."

Key Points

- Used for testing a population mean (μ) when population standard deviation (σ) is known.
- Requires independent and random samples.
- Works well for large sample sizes (typically $n \geq 30$).
- Based on the standard normal distribution (Z-distribution).
- Helps in decision-making using sample data.
- Widely used in business, research, operations, and quality control.

Why is it Important?

- Helps validate business claims about an average value.
- Used to check process performance (e.g., delivery time, production time).
- Supports A/B testing and campaign effectiveness analysis.
- Helps in quality control and manufacturing.
- Enables data-driven decisions before making changes.
- Reduces risk of wrong decisions based only on sample data.

Z-Test Formula

$$Z = \frac{\bar{X} - \mu_0}{\sigma \sqrt{n}}$$

$\bar{X}$	Sample mean
μ_0	Hypothesized population mean
σ	Population standard deviation (known)
n	Sample size

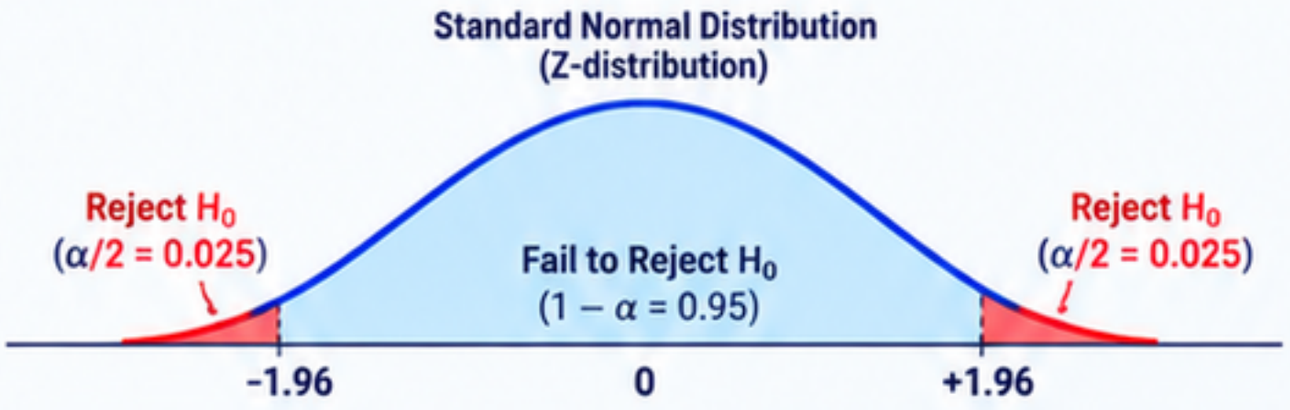
When to Use Z-Test?

- Population standard deviation (σ) is known.
- Sample is random and independent.
- Sample size is large ($n \geq 30$).
- Data is approximately normally distributed (or n is large due to Central Limit Theorem).

When NOT to Use Z-Test?

- Population standard deviation (σ) is unknown (use t-test instead).
- Small sample size ($n < 30$) with unknown σ.
- Data is not approximately normal (for small samples).
- Samples are dependent (use paired t-test instead).

Visual Representation



For α = 0.05 (Two-tailed test)

- Critical value on left: −1.96
- Critical value on right: +1.96
- If $Z < -1.96$ or $Z > +1.96$ → Reject H_0
- If $-1.96 \leq Z \leq +1.96$ → Fail to reject H_0

Real-World Business Examples

Business Area	Example Question	Why Use Z-Test?
E-commerce	Is the average delivery time 30 minutes?	Population standard deviation is known from historical data.
Marketing	Is the average order value ₹1,000?	Large sample size and known population σ.
Manufacturing	Is the average product weight 500 grams?	σ is known from process specifications.
Finance	Is the average transaction value different from ₹5,000?	Large transaction history, known σ.
Healthcare	Is the average recovery time 7 days?	σ is known and sample size is large.

Excel Calculation Example

A company claims the average delivery time is 30 minutes. A sample of 36 deliveries gives a mean of 32 minutes. Population σ = 6 minutes. Test at α = 0.05.

Parameter	Value
Hypothesized mean (μ_0)	30
Sample mean ($\bar{X}$)	32
Population σ	6
Sample size (n)	36
Significance level (α)	0.05

Excel Formulas		Value
Standard Error	=B4/SQRT(B5)	1
Z-statistic	=(B3-B2)/B8	2
P-value (two-tailed)	=2*(1-NORM.S.DIST(ABS(B9),TRUE))	0.0455

Result: p-value = 0.0455 < 0.05 → Reject H_0

Interpretation

Statistical Conclusion
There is statistically significant evidence that the average delivery time is different from 30 minutes at the 5% significance level.

Business Interpretation
The observed increase to 32 minutes is not likely due to random chance. The company should investigate the reasons and consider process improvements.

Key Takeaways

- Use Z-test when population σ is known.
- Requires large sample size ($n \geq 30$).
- Helps validate claims about a mean.
- Compare p-value or critical value.
- Always interpret results in business context, not just statistically.

2 One Sample Z-Test

Testing a Single Population Mean (σ Known)

What is One Sample Z-Test?

A one sample Z-test is used to check whether the sample mean is significantly different from a known population mean (μ_0) when the population standard deviation (σ) is known.

> "It helps us test a claim about a single population mean using sample data."

Key Points

- Used to test a claim about a single population mean.
- Population standard deviation (σ) is known.
- Sample is random and independent.
- Appropriate for large samples (typically n ≥ 30).
- Based on standard normal distribution (Z-distribution).
- Can be one-tailed or two-tailed.
- Widely used in business, quality control, marketing, and operations.

Hypotheses for One Sample Z-Test

- **Null Hypothesis (H_0):** States the current claim or assumption. Example: $H_0 : \mu = 30$
- **Alternative Hypothesis (H_1):** Represents the effect or difference we want to test.

Forms of H_1:

- **Two-tailed:** $H_1 : \mu \neq 30$ (Check for any difference)
- **Right-tailed:** $H_1 : \mu > 30$ (Check for increase)
- **Left-tailed:** $H_1 : \mu < 30$ (Check for decrease)

Test Statistic Formula

$$Z = \frac{\bar{X} - \mu_0}{\sigma \sqrt{n}}$$

$\bar{X}$	Sample mean
μ_0	Hypothesized population mean
σ	Population standard deviation (known)
n	Sample size

Steps in One Sample Z-Test

1. State the problem (business question).
2. Formulate H_0 and H_1.
3. Choose significance level (α).
4. Calculate the test statistic (Z).
5. Find the p-value (or compare with critical value).
6. Make a decision (Reject or Fail to Reject H_0).
7. Interpret the result in business context.

Types of One Sample Z-Test

Two-tailed Test	$H_1 : \mu \neq \mu_0$	Checks for any difference (increase or decrease).
Right-tailed Test	$H_1 : \mu > \mu_0$	Checks for an increase in mean.
Left-tailed Test	$H_1 : \mu < \mu_0$	Checks for a decrease in mean.

Real-World Example: Delivery Time

A company claims that the average delivery time is 30 minutes.
A sample of 36 deliveries gives a mean delivery time of 32 minutes.
Population standard deviation is known to be 6 minutes.
Test at α = 0.05 (two-tailed test).

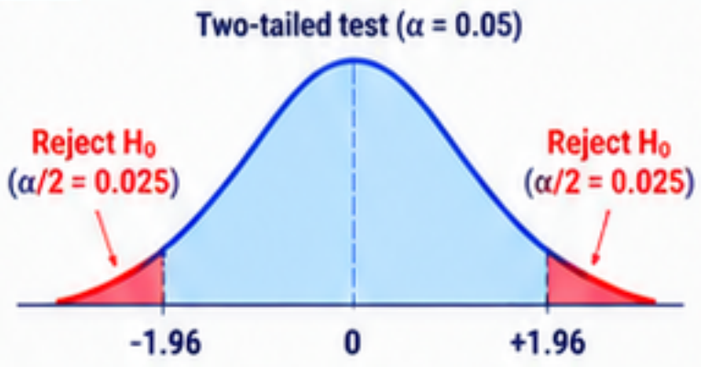
Parameter	Value
Hypothesized mean (μ_0)	30
Sample mean ($\bar{X}$)	32
Population standard deviation (σ)	6
Sample size (n)	36
Significance level (α)	0.05

Excel Calculation

Calculation Step	Excel Formula	Result
Standard Error (SE)	=B4/SQRT(B5)	1
Z-statistic	=(B3-B2)/B6	2
P-value (two-tailed)	=2*(1-NORM.S.DIST (ABS(B7),TRUE))	0.0455
Decision	=IF(B8<=B5, "Reject H0", "Fail to Reject H0")	Reject H_0

p-value = 0.0455 < 0.05
Reject H_0

Normal Distribution & Decision



Since Z = 2 is greater than +1.96 (or p-value = 0.0455 < 0.05), we **reject H_0**. There is statistically significant evidence that the average delivery time is different from 30 minutes.

Business Use Cases

- **E-commerce:** Is the average delivery time 30 minutes?
- **Marketing:** Is the average order value ₹1,000?
- **Manufacturing:** Is the average product weight 500 grams?
- **Finance:** Is the average transaction value different from ₹5,000?
- **HR Analytics:** Is the average employee satisfaction higher than 7?
- **Healthcare:** Is the average recovery time 7 days?

Interpretation in Business Context

- The average delivery time is significantly different from 30 minutes.
- Since the sample mean (32) is higher, it indicates an increase in delivery time.
- The company should investigate reasons (e.g., operational delays, increased demand).
- Consider process improvements to reduce delivery time and improve customer satisfaction.

Key Takeaways

- Use one sample Z-test when population σ is known.
- Follow a step-by-step process.
- Compare p-value with α or use critical value.
- Interpret results in business terms.
- Widely used in real-world scenarios.
- Helps in evidence-based decision-making.
- Understand the direction and size of the effect, not just statistical significance.

3 Two Sample Z-Test

Comparing Two Independent Population Means (σ Known)

What is Two Sample Z-Test?

A two sample Z-test is used to compare the means of two independent populations when the population standard deviations (σ_1 and σ_2) are known. It helps us check whether the difference between two sample means is statistically significant or just due to random chance.

> "It helps us compare the performance of two groups and decide if the observed difference is real."

Key Points

- Used to compare the means of two independent groups.
- Population standard deviations (σ_1 and σ_2) are known.
- Samples are independent (no overlap).
- Based on standard normal distribution (Z-distribution).
- Works well for large sample sizes (typically $n_1 \geq 30$ and $n_2 \geq 30$).
- Can be one-tailed or two-tailed.
- Commonly used in business, A/B testing, marketing, HR analytics, and manufacturing.

Hypotheses for Two Sample Z-Test

Null Hypothesis (H_0):
There is no difference between the population means.
$H_0 : \mu_1 - \mu_2 = 0$ (or $\mu_1 = \mu_2$)

Alternative Hypothesis (H_1):
There is a difference between the population means.

Forms of H_1:

- **Two-tailed:** $H_1 : \mu_1 - \mu_2 \neq 0$ (Check for any difference)
- **Right-tailed:** $H_1 : \mu_1 - \mu_2 > 0$ (Check if group 1 mean is higher)
- **Left-tailed:** $H_1 : \mu_1 - \mu_2 < 0$ (Check if group 1 mean is lower)

Test Statistic Formula

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$\bar{X}_1$	Sample mean of group 1
$\bar{X}_2$	Sample mean of group 2
σ_1	Population standard deviation of group 1 (known)
σ_2	Population standard deviation of group 2 (known)
n_1	Sample size of group 1
n_2	Sample size of group 2

Steps in Two Sample Z-Test

1. State the problem (business question).
2. Formulate H_0 and H_1.
3. Choose significance level (α).
4. Calculate the test statistic (Z).
5. Find the p-value (or compare with critical value).
6. Make a decision (Reject or Fail to Reject H_0).
7. Interpret the result in business context.

Types of Two Sample Z-Test

Two-tailed Test	$H_1 : \mu_1 - \mu_2 \neq 0$	Checks if there is any difference between group means.
Right-tailed Test	$H_1 : \mu_1 - \mu_2 > 0$	Checks if group 1 mean is higher than group 2.
Left-tailed Test	$H_1 : \mu_1 - \mu_2 < 0$	Checks if group 1 mean is lower than group 2.

Real-World Example: Marketing Campaign

A company runs two different marketing campaigns (A and B) to increase website conversion rate. They want to know if the new campaign (B) is performing significantly better than the existing campaign (A). Population standard deviations are known from historical data.

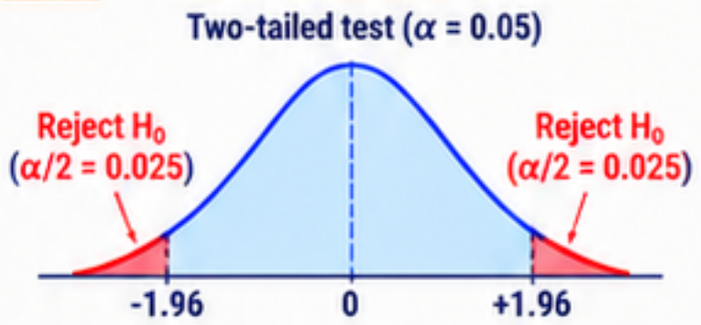
Parameter	Campaign A (Old)	Campaign B (New)
Sample mean ($\bar{X}$)	5.0%	6.2%
Population σ (known)	2.0%	2.5%
Sample size (n)	100	120
Significance level (α)		0.05

Excel Calculation

Step	Excel Formula	Result
Standard Error (SE)	=SQRT((B3^2/B4)+(C3^2/C4))	0.3202
Z-statistic	=(B2-C2)/B5	-3.745
P-value (two-tailed)	=2*(1-NORM.S.DIST(ABS(B6),TRUE))	0.00018
Decision	=IF(B7<=B5,"Reject H_0","Fail to Reject H_0")	Reject H_0

p-value = 0.00018 < 0.05
Reject H_0

Normal Distribution & Decision



Since Z = -3.745 is less than -1.96 and p-value = 0.00018 < 0.05, we reject H_0. There is statistically significant evidence that the new campaign has a different conversion rate than the old campaign.

Business Use Cases

- **E-commerce:** Compare conversion rates between two designs.
- **Marketing:** Compare campaign performance (A vs B).
- **Manufacturing:** Compare average product from two machines.
- **HR Analytics:** Compare average employee satisfaction between two segments.
- **Finance:** Compare average transaction value between two treatments.
- **Healthcare:** Compare average recovery time between two treatments.

Interpretation in Business Context

- The new campaign has a significantly different conversion rate than the old campaign.
- Since the new campaign mean (6.2%) is higher, it indicates better performance.
- The company can consider rolling out the new campaign to a wider audience.
- Also evaluate the actual business impact (cost, ROI, scalability) before final decision.

Key Takeaways

- Use two sample Z-test when population σ_1 and σ_2 are known.
- Compare means of two independent groups.
- Follow a step-by-step process.
- Can be one-tailed or two-tailed.
- Interpret results in business context.
- Widely used in A/B testing, marketing, operations, and HR analytics.
- Always check assumptions and sample size.

4 Assumptions of Z-Test

Conditions for Valid and Reliable Results

What are Assumptions?

Assumptions are the conditions that must be satisfied for the Z-test to give valid and reliable results.
If these assumptions are not met, the p-value and test conclusions may not be accurate.

> "Always check assumptions before performing a statistical test, otherwise the results can be misleading."

Key Assumptions of Z-Test (When to Use)

1. Population standard deviation (σ) is known.
2. The sample is random and independent.
3. The population is approximately normally distributed (or sample size is large, typically $n \geq 30$).
4. For two sample Z-test: the two samples are independent.
5. Observations within each sample are independent of each other.

Why These Assumptions Matter?

- Ensure the sampling distribution of the sample mean is normal.
- Help in accurate calculation of the standard error and test statistic.
- Reduce the risk of incorrect conclusions.
- Make the p-value and critical values reliable.
- Especially important for small sample sizes ($n < 30$).

1 Population Standard Deviation is Known

Z-test requires the population standard deviation (σ) to be known.
If σ is unknown, we use the t-test instead.

Example:
A company knows from historical data that the population standard deviation of delivery time is 6 minutes.

How to get σ?

- From historical data
- From industry reports
- From process specifications
- From complete population data (rare)

2 Sample is Random and Independent

The sample should be selected randomly from the population, and each observation should be independent of the others.

Example:
Selecting 36 delivery orders randomly from all orders ensures a valid test.

3 Population is Approximately Normal (or Large Sample Size)

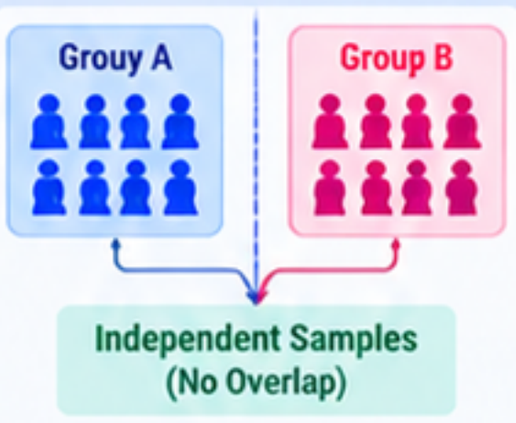
The population should be normally distributed, or the sample size should be large ($n \geq 30$).
Due to the Central Limit Theorem, the sampling distribution of the mean becomes approximately normal for large sample sizes, even if the original population is not normal.

Visual Explanation

4 Two Samples are Independent (For Two Sample Z-Test)

The two samples should be independent of each other. There should be no overlap between the observations in the two groups.

Example:
Campaign A and Campaign B are tested on two different groups of users. No user appears in both groups.



5 Observations are Independent

Each observation within a sample should be independent.
One value should not influence another.

Example:
Each delivery order is placed by a different customer.
The delivery time of one order does not affect another order.

Violation Example:
Measuring temperature of the same machine multiple times can lead to dependent observations.

How to Check Assumptions in Practice?

Assumption	How to Check?	Tools / Methods
σ is known	Check if available from historical or industry data	Business records, domain knowledge
Random sample	Check sampling method	Data collection process
Independence	Ensure no overlap or influence between samples	Data design, domain logic
Normal distribution (or n ≥ 30)	Plot histogram / Q-Q plot or check sample size	Excel histogram, Q-Q plot, n ≥ 30 rule

What if Assumptions are Violated?

Violation	What to Do?
Population σ is unknown	Use t-test instead
Small sample size (n < 30) and non-normal data	Use t-test or non-parametric test
Samples are dependent	Use paired t-test instead of two sample Z-test
Data is heavily skewed with small sample	Use data transformation or non-parametric test

Using Z-test when assumptions are not met can lead to incorrect p-values and wrong business decisions.

Key Takeaways

- Z-test requires population σ to be known.
- Use random and independent samples.
- Population should be normal or sample size should be large ($n \geq 30$).
- Two samples must be independent for two sample Z-test.
- Check assumptions using data visualization and domain knowledge.
- If assumptions are violated, use t-test or other appropriate methods.
- Valid assumptions → Reliable results → Better business decisions.

5 T-Test: When and Why?

Testing a Population Mean When Standard Deviation is Unknown

What is a T-Test?

A t-test is a statistical test used to check whether a sample mean is significantly different from a hypothesized population mean when the population standard deviation (σ) is unknown.

> "We use t-test when we don't know the population standard deviation and we have to estimate it from the sample data."

Key Points

- Used when population standard deviation (σ) is unknown.
- Uses sample standard deviation (s).
- Follows a t-distribution (not normal distribution).
- Suitable for small samples (typically n < 30).
- Can also be used for large samples (n ≥ 30).
- We will cover one-sample, two-sample (independent) and paired t-tests.
- Widely used in business, research, marketing, healthcare and operations.

Why is it Important?

- Helps validate business claims when σ is not available.
- Used in real-world situations where we only have sample data.
- Provides more reliable results for small sample sizes.
- Widely used in market research, quality control, HR analytics, and A/B testing.
- Helps in data-driven decision making with limited data.
- More flexible than Z-test as it works even when σ is unknown.

T-Test Formula (One Sample)

$$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

$\bar{X}$	Sample mean
μ_0	Hypothesized population mean
s	Sample standard deviation
n	Sample size
t	Test statistic (follows t-distribution)
df	Degrees of freedom $(n - 1)$

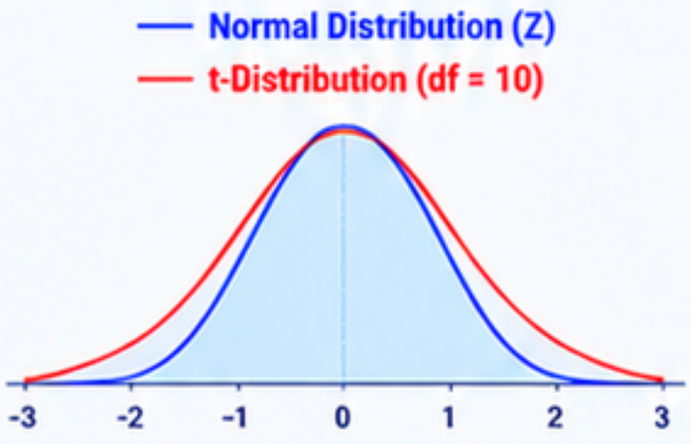
When to Use T-Test?

- Population standard deviation (σ) is unknown.
- Sample is random and independent.
- Sample size is small (n < 30) or moderate.
- Data is approximately normally distributed (or no strong evidence against normality).
- For two-sample t-test, the two samples are independent.
- For paired t-test, the data is before-after or matched pairs.

Types of T-Test

One Sample T-Test	Tests whether the mean of a single population is different from a given value (σ unknown). $H_1: \mu \neq \mu_0,\ \mu > \mu_0$ or $\mu < \mu_0$
Independent Two Sample T-Test	Compares the means of two independent groups. $H_1: \mu_1 \neq \mu_2,\ \mu_1 > \mu_2$ or $\mu_1 < \mu_0$
Paired Sample T-Test	Compares the means of two related groups (before-after). $H_1: \mu_d \neq 0,\ \mu_d > 0$ or $\mu_d < 0$

T-Distribution vs Normal Distribution



Key Differences

- t-distribution has heavier tails than normal distribution.
- As sample size increases, t-distribution approaches normal distribution.
- For small samples (n < 30), t-distribution is preferred.
- Degrees of freedom (df) = $n - 1$.

Real-World Example: Customer Satisfaction

A company wants to check if the average customer satisfaction score is different from 7. The population standard deviation is unknown. A sample of 16 customers gives a mean score of 7.8 with a sample standard deviation of 1.5. Test at $\alpha = 0.05$ (two-tailed test).

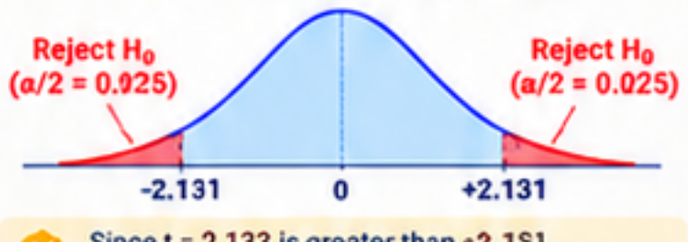
Business Question
Is the average customer satisfaction significantly different from ??

Parameter	Value
Hypothesized mean (μ_0)	7
Sample mean ($\bar{X}$)	7.8
Sample standard deviation (s)	1.5
Sample size (n)	16
Degrees of freedom (df = n − 1)	15
Significance level (α)	0.05

Excel Calculation

Step	Excel Formula	Result
Standard Error (SE)	=B4/SQRT(B5)	0.375
t-statistic	=(B3-B2)/B6	2.133
p-value (two-tailed)	=2*(1-T.DIST.ABS(B7,B5-1),TRUE)	0.0497
Decision	=IF(B8<=B6, "Reject H_0", "Fail to Reject H_0")	Reject H_0

Critical Value & Decision



Since t = 2.133 is greater than +2.131 and p-value = 0.0497 < 0.05, we reject H_0. We accept, "The customer satisfaction score is significantly different from 7.

Business Interpretation

- The sample provides statistically significant evidence that the average customer satisfaction score is different from 7.
- Since the mean score is higher (7.8), it indicates an improvement in customer satisfaction.
- The company can analyze the factors contributing to this improvement and continue the current strategy.

Business Use Cases

E-commerce	Marketing	Manufacturing	HR Analytics	Finance	Healthcare
Customer satisfaction scores	Campaign effectiveness (before-after)	Product quality measurements	Employee satisfaction scores	Average transaction value	Treatment recovery time

Key Takeaways

- Use t-test when population standard deviation is unknown.
- Works well for small sample sizes (n < 30).
- Based on t-distribution with degrees of freedom.
- Can be one-sample, independent two-sample or paired t-test.
- Helps in real-world decision making with limited data.

6 One Sample T-Test

Testing a Population Mean When Standard Deviation is Unknown

What is One Sample T-Test?

A one sample t-test is used to check whether the sample mean is significantly different from a hypothesized population mean (μ_0) when the population standard deviation (σ) is unknown.

> "We use a t-test when we don't know the population standard deviation and we estimate it from the sample data."

Key Points

- Used to test a claim about a single population mean.
- Population standard deviation (σ) is unknown.
- Uses sample standard deviation (s).
- Follows a t-distribution with $(n-1)$ degrees of freedom.
- Suitable for small samples (typically $n < 30$).
- Can also be used for large samples ($n \geq 30$).
- Can be one-tailed or two-tailed.
- Widely used in business, research, quality control, and operations.

Hypotheses for One Sample T-Test

- **Null Hypothesis (H_0):** States the current claim or assumption. Example: $H_0 : \mu = 50$
- **Alternative Hypothesis (H_1):** Represents the effect or difference we want to test.

Forms of H_1:

- **Two-tailed:** $H_1 : \mu \neq 50$ (Check for any difference)
- **Right-tailed:** $H_1 : \mu > 50$ (Check for increase)
- **Left-tailed:** $H_1 : \mu < 50$ (Check for decrease)

T-Test Formula (One Sample)

$$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

$\bar{X}$	Sample mean
μ_0	Hypothesized population mean
s	Sample standard deviation
n	Sample size
t	Test statistic (follows t-distribution)
df	Degrees of freedom $(n-1)$

Steps in One Sample T-Test

1. State the problem (business question).
2. Formulate H_0 and H_1.
3. Choose significance level (α).
4. Calculate the test statistic (t).
5. Find the p-value (or compare with critical value).
6. Make a decision (Reject or Fail to Reject H_0).
7. Interpret the result in business context.

Assumptions

- Population standard deviation (σ) is unknown.
- Sample is random and independent.
- Data is approximately normally distributed (or sample size is large: $n \geq 30$).
- No extreme outliers in the data.
- For small samples ($n < 30$), normality is important.

Real-World Example: Average Call Handling Time

A call center claims that the average call handling time is 5 minutes. A sample of 12 calls gives a mean handling time of 5.8 minutes with a sample standard deviation of 1.5 minutes.
Test at $\alpha = 0.05$ (two-tailed test).

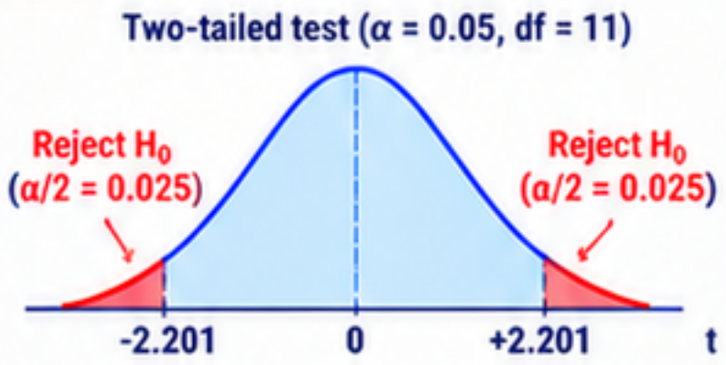
Parameter	Value
Hypothesized mean (μ_0)	5
Sample mean ($\bar{X}$)	5.8
Sample standard deviation (s)	1.5
Sample size (n)	12
Degrees of freedom ($df = n-1$)	11
Significance level (α)	0.05

Excel Calculation

Calculation Step	Excel Formula	Result
Standard Error (SE)	=B4/SQRT(B5)	0.4330
t-statistic	=(B3-B2)/B6	1.8475
p-value (two-tailed)	=2*(1-T.DIST.ABS(B7),B8,TRUE)	0.0938
Decision	=IF(B9<=B10, "Reject H_0", "Fail to Reject H_0")	Fail to Reject H_0

p-value = 0.0938 > 0.05
Fail to Reject H_0

t-Distribution & Decision



Since t = 1.8475 is between -2.201 and +2.201 (and p-value = 0.0938 > 0.05), we fail to reject H_0. There is not sufficient evidence that the average call handling time is different from 5 minutes.

Business Use Cases

- **Customer Support:** Average handling time, resolution time
- **E-commerce:** Average order, delivery time
- **Manufacturing:** Average product weight, defect rate
- **HR Analytics:** Average training score, employee satisfaction
- **Finance:** Average transaction value
- **Healthcare:** Average recovery time, patient stay duration

Interpretation in Business Context

- The sample mean (5.8) is higher than the claim (5), but the difference is not statistically significant.
- The company cannot conclude that the average handling time is different from 5 minutes.
- They can continue monitoring or collect more data to get a clearer result.

Key Takeaways

- Use one sample t-test when σ is unknown.
- Based on t-distribution with $(n-1)$ df.
- Suitable for small samples ($n < 30$).
- Compare p-value with α or use critical value.
- Check assumptions (normality, no outliers).
- Interpret results in business context.
- Widely used in real-world scenarios.

7 Independent Two Sample T-Test

Comparing the Means of Two Independent Groups

What is Independent Two Sample T-Test?

An independent two sample t-test is used to compare the means of two independent groups to check whether there is a statistically significant difference between them, when the population standard deviations are unknown.

> "We use an independent two sample t-test when we want to compare the average of two different groups."

Key Points

- Used to compare the means of two independent groups.
- Population standard deviations (σ_1 and σ_2) are unknown.
- Samples are independent (no overlap).
- Follows a t-distribution.
- Can be one-tailed or two-tailed.
- Suitable for small and moderate sample sizes.
- Commonly used in business, A/B testing, HR analytics, marketing, and quality control.

Hypotheses for Independent Two Sample T-Test

Null Hypothesis (H_0):
There is no significant difference between the population means.
$H_0 : \mu_1 - \mu_2 = 0$ (or $\mu_1 = \mu_2$)

Alternative Hypothesis (H_1):
There is a significant difference between the population means.

Forms of H_1:

- **Two-tailed:** $H_1 : \mu_1 - \mu_2 \neq 0$ (Check for any difference)
- **Right-tailed:** $H_1 : \mu_1 - \mu_2 > 0$ (Check if group 1 mean is higher)
- **Left-tailed:** $H_1 : \mu_1 - \mu_2 < 0$ (Check if group 1 mean is lower)

Test Statistic Formula

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{X}_1$	Sample mean of group 1
$\bar{X}_2$	Sample mean of group 2
s_1	Sample standard deviation of group 1
s_2	Sample standard deviation of group 2
n_1	Sample size of group 1
n_2	Sample size of group 2
t	Test statistic (follows t-distribution)

Steps in Independent Two Sample T-Test

1. State the problem (business question).
2. Formulate H_0 and H_1.
3. Choose significance level (α).
4. Check assumptions (independence, normality, etc.).
5. Calculate the test statistic (t).
6. Find the p-value (or compare with critical value).
7. Make a decision (Reject or Fail to Reject H_0).
7. Interpret the result in business context.

Assumptions

- The two samples are independent.
- Population standard deviations are unknown.
- Data in each group is approximately normally distributed (or sample size is large: $n \geq 30$).
- No extreme outliers.
- Homogeneity of variances (equal variances) or use Welch's t-test if variances are unequal.

Real-World Example: Training Program Effectiveness

A company wants to check if a new training program improves employee productivity. Two independent groups of employees are considered: Group A (without training) and Group B (with training).

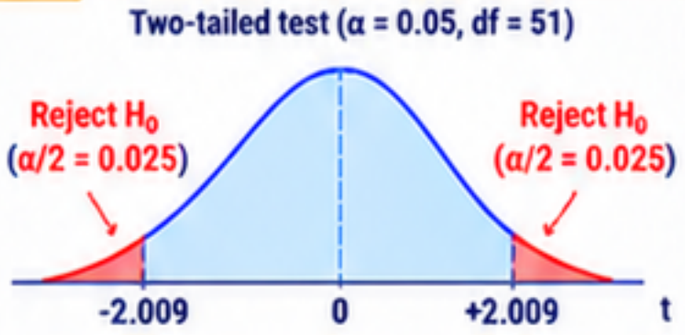
Parameter	Group A (No Training)	Group B (With Training)
Sample size (n)	25	28
Sample mean ($\bar{X}$)	70	78
Sample standard deviation (s)	10	12
Significance level (α)	0.05	

Excel Calculation

Calculation Step	Excel Formula	Result
Standard Error (SE)	=SQRT((B3^2/B2)+(C3^2/C2))	3.121
t-statistic	=(B4-C4)/B5	-2.564
Degrees of dfreedom (df)	=B2+C2-2	51
p-value (two-tailed)	=2*(1-T.DIST.ABS(B6),B7,TRUE)	0.0134
Decision	=IF(B8<=0.05, "Reject H_0", "Fail to Reject H_0")	Reject H_0

p-value = 0.0134 < 0.05
Reject H_0

t-Distribution & Decision



Since |t| = 2.564 is greater than 2.009 and p-value = 0.0134 < 0.05, we reject H_0. There is statistically significant evidence that the training program has improved employee productivity.

Business Use Cases

Marketing
Compare campaign performance (A vs B).

HR Analytics
Compare training program effectiveness.

E-commerce
Compare average order values between two groups.

Manufacturing
Compare defect rates between two machines or shifts.

Healthcare
Compare recovery time between two treatments.

Interpretation in Business Context

- The training program group has a higher average productivity (78 vs 70).
- The difference is statistically significant ($p < 0.05$).
- The company can continue and expand the training program.
- Also evaluate the practical impact (e.g., cost, ROI, long-term benefits).

Key Takeaways

- Independent two sample t-test compares means of two independent groups.
- Use when population standard deviations are unknown.
- Check assumptions (independence, normality, equal variances).
- Can be one-tailed or two-tailed.
- Interpret results in business context.
- Widely used in A/B testing, marketing, HR, and operations.

8 Paired Sample T-Test

Comparing Means of Two Related Samples (Before-After Analysis)

What is a Paired Sample T-Test?

A paired sample t-test is used to compare the means of two related samples to check whether there is a statistically significant difference between them. The two samples are dependent, meaning the observations are related or matched (e.g., before and after measurements on the same group of people).

> "We use a paired t-test when we want to measure the impact of a change on the same group of individuals."

Key Points

- Used to compare means of two related groups (before-after).
- Observations are dependent (paired).
- Measures the mean difference between pairs.
- Follows a t-distribution with $(n-1)$ degrees of freedom.
- Suitable for small and moderate sample sizes.
- Commonly used in business, healthcare, education, and research to evaluate the impact of an intervention or change.

Hypotheses for Paired Sample T-Test

Null Hypothesis (H_0):
There is no significant difference between the paired means (mean difference = 0).
$H_0 : \mu_d = 0$

Alternative Hypothesis (H_1):
There is a significant difference between the paired means.
$H_1 : \mu_d \neq 0$ (or > 0 or < 0)

Forms of H_1:

- **Two-tailed:** $H_1 : \mu_d \neq 0$ (Check for any difference)
- **Right-tailed:** $H_1 : \mu_d > 0$ (Check if mean increased)
- **Left-tailed:** $H_1 : \mu_d < 0$ (Check if mean decreased)

Test Statistic Formula

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

$\bar{d}$	Mean of the differences (d = Before − After)
s_d	Standard deviation of the differences
n	Number of paired observations
t	Test statistic (follows t-distribution)
df	Degrees of freedom $(n-1)$
d	Difference for each pair (Before − After)

Steps in Paired Sample T-Test

1. State the problem (business question).
2. Compute the difference for each pair (d = Before − After).
3. Formulate H_0 and H_1.
4. Choose significance level (α).
5. Calculate mean of differences ($\bar{d}$) and standard deviation (s_d).
6. Compute the test statistic (t).
7. Find the p-value (or compare with critical value).
8. Make a decision (Reject or Fail to Reject H_0).
9. Interpret the result in business context.

Assumptions

- The pairs are dependent (same individuals or matched pairs).
- The differences (d) are approximately normally distributed.
- No extreme outliers in the differences.
- Data is measured on an interval or ratio scale.
- Suitable for small and moderate sample sizes ($n < 30$ or $n \geq 30$).

Real-World Example: Sales Training Impact

A company conducts a sales training program and wants to check if the training significantly improves the sales performance of the same 10 employees. We compare their monthly sales (before and after training).

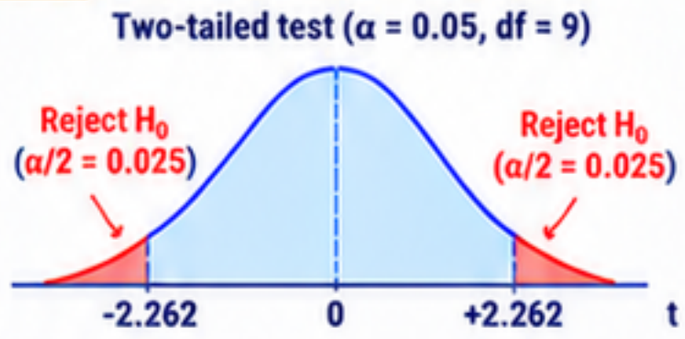
Employee	Before Training	After Training	Difference (d) (After − Before)
E1	50	58	8
E2	42	51	9
E3	60	65	5
E4	55	63	8
E5	48	56	8
E6	72	80	8
E7	40	47	7
E8	65	70	5
E9	58	66	8
E10	47	55	8

Excel Calculation

Calculation Step	Excel Formula	Result
Differences (d)	=C2-B2	-
Mean of differences ($\bar{d}$)	=AVERAGE(D2:D11)	7.0
Standard deviation (s_d)	=STDEV.S(D2:D11)	1.41
Sample size (n)	=COUNT(D2:D11)	10
t-statistic	=$\bar{D}$/(s_d/SQRT(n))	15.72
Degrees of freedom (df)	=B10-1	9
p-value (two-tailed)	=2*(1-T.DIST.ABS(15.72,9))	0.00001
Decision	=IF(B7<0.05,"Reject H_0","Fail to Reject H_0")	Reject H_0

p-value = 0.00001 < 0.05
Reject H_0

t-Distribution & Decision



Since |t| = 15.72 is greater than 2.262 and p-value = 0.00001 < 0.05, we reject H_0. There is statistically significant evidence that the sales training program has improved employee sales performance.

Business Use Cases

Marketing
Measure campaign impact (before vs after campaign)

Sales
Compare sales performance (before-after training)

Operations
Evaluate process improvements (before-after implementation)

HR Analytics
Measure employee performance (before-after training)

Healthcare
Compare patient outcomes (before-after treatment)

Interpretation in Business Context

- The average sales increased by 7 units after training.
- The improvement is statistically significant ($p < 0.05$).
- The training program is effective and can be rolled out to more employees.
- Consider the cost, scalability, and long-term impact before final decision.

Key Takeaways

- Use paired sample t-test for related samples.
- Analyze the differences between pairs.
- Ensure differences are approximately normally distributed.
- Suitable for small and moderate samples.
- Widely used in before-after analysis.
- Helps evaluate the impact of a change or intervention.

9 Assumptions of T-Test

Conditions for Valid and Reliable Results

What are Assumptions?

Assumptions are the conditions that must be satisfied for the t-test to give valid and reliable results. If these assumptions are not met, the p-value, confidence intervals and test conclusions may be inaccurate.

> "Assumptions ensure that the t-test results are trustworthy and meaningful in real-world analysis."

Key Assumptions of T-Test

1. Population standard deviation (σ) is unknown.
2. The sample is randomly selected.
3. The observations are independent.
4. The data is approximately normally distributed (or sample size is large: $n \geq 30$).
5. No extreme outliers in the data.
6. For two-sample t-test: the two samples are independent (unpaired) or correctly paired.

Why These Assumptions Matter?

- Ensures accurate calculation of standard error and test statistic.
- Keeps the t-distribution valid.
- Reduces the risk of incorrect conclusions.
- Makes p-values and confidence intervals reliable.
- Helps in better business decision making.
- Especially important for small sample sizes ($n < 30$).

4 Population Standard Deviation is Unknown

T-test is used when the population standard deviation (σ) is unknown. We use the sample standard deviation (s) to estimate it.

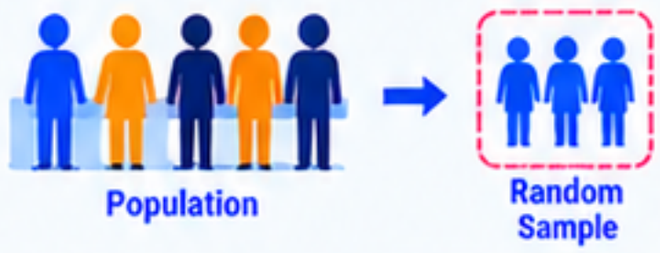
Example (Business Context):
A company wants to test if the average delivery time is 30 minutes. The population standard deviation is unknown, so we use the sample standard deviation from historical data.

- In most real-world scenarios, we do not have complete population data. Hence, we rely on sample data to estimate the variability.

5 Sample is Random and Independent

The sample should be selected randomly from the population and each observation should be independent of the others.

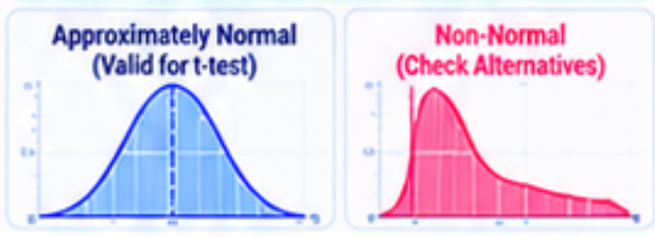
Example:
A retail store randomly selects 20 customers to measure their billing time. The selection should not be biased (e.g., only selecting loyal customers).



6 Data is Approximately Normally Distributed

The t-test assumes that the data follows a normal distribution, especially for small sample sizes ($n < 30$). For large samples ($n \geq 30$), the t-test is robust to moderate deviations from normality due to the Central Limit Theorem.

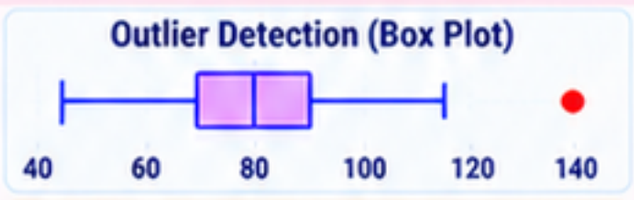
Example:
A company collects 25 product weights. We can check the distribution using a histogram or Q-Q plot to ensure it is approximately normal.



7 No Extreme Outliers

T-test is sensitive to extreme outliers, especially for small sample sizes. Outliers can affect the mean and standard deviation, leading to incorrect results.

Example:
If most employees have a training score around 70, but one employee has a score of 150, it can heavily influence the results.



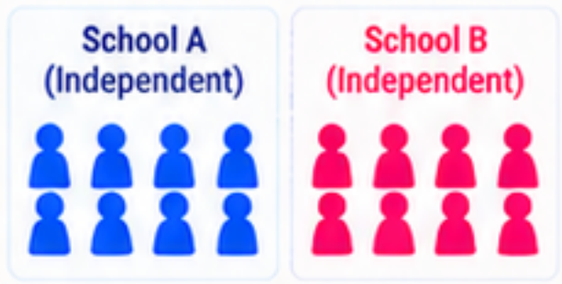
Methods to Detect Outliers:

- Box plot
- Z-score ($|Z| > 3$)
- IQR method ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$)
- Visual inspection (scatter/histogram)

8 Independence Between Two Samples (For Independent Two Sample T-Test)

In a two-sample t-test (independent), the observations in one group should not be related to the observations in the other group.

Example:
Comparing test scores of students from two different schools. A student from School A should not be the same as a student from School B.



9 How to Check Assumptions in Excel?

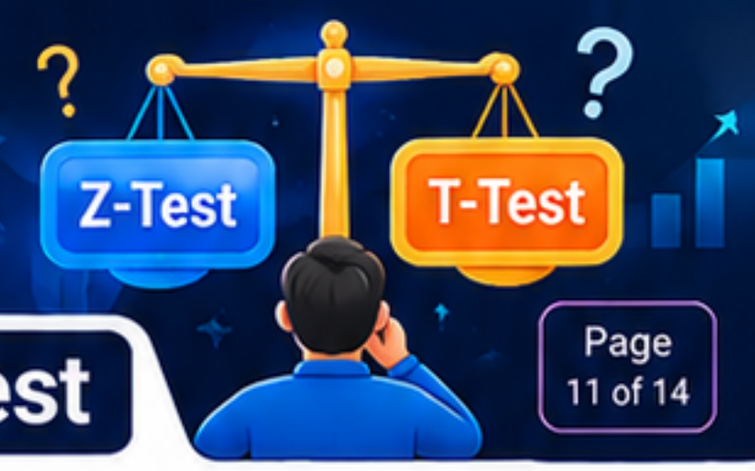
Assumption	How to Check?	Excel Tools / Functions
σ is unknown	Use sample standard deviation (s)	STDEV.S(range)
Random sample	Ensure unbiased selection	Sampling method, data collection process
Independence	Check no relationship between observations	Study design, domain knowledge
Normality	Histogram, Q-Q plot, Shapiro-Wilk test	Analysis ToolPak (Descriptive Statistics), Q-Q Plot (add-in)
Outliers	Box plot, Z-score, IQR	Box & Whisker Chart, conditional formatting, formulas
Two-sample independence	Ensure groups are unrelated	Study design, business logic

10 What if Assumptions are Violated?

Violation	What to Do?
Strong non-normality (small sample)	Use non-parametric test (e.g., Wilcoxon Signed-Rank or Mann-Whitney).
Presence of outliers	Remove or treat outliers, or use robust methods.
Dependent samples (when independent required)	Use paired t-test instead.
Unequal variances (for two-sample t-test)	Use Welch's t-test.
Small sample with severe non-normality	Use non-parametric test or transform the data (e.g., log transformation).

11 Key Takeaways

- T-test assumes population standard deviation is unknown.
- Sample should be random and independent.
- Data should be approximately normally distributed ($n \geq 30$).
- No extreme outliers.
- Two samples should be independent (for independent t-test).
- Always check assumptions before performing a t-test.
- If assumptions are violated, consider alternative methods.



10 Choosing Between Z-Test and T-Test

A Practical Guide for Real-World Decision Making

Why Do We Need to Choose?

Both Z-test and T-test are used to test hypotheses about population means. Choosing the correct test is important because it directly affects the accuracy of the results. Using the wrong test can lead to incorrect p-values and wrong business conclusions.

> "The right test depends on what information you have (population standard deviation) and the sample size."

Key Factors for Choosing

1. Is the population standard deviation (σ) known or unknown?
2. What is the sample size (n)?
3. Is the data approximately normally distributed?
4. Are you comparing one sample or two samples?
5. Are the samples independent or paired (related)?
6. Are there any outliers or violations of assumptions?

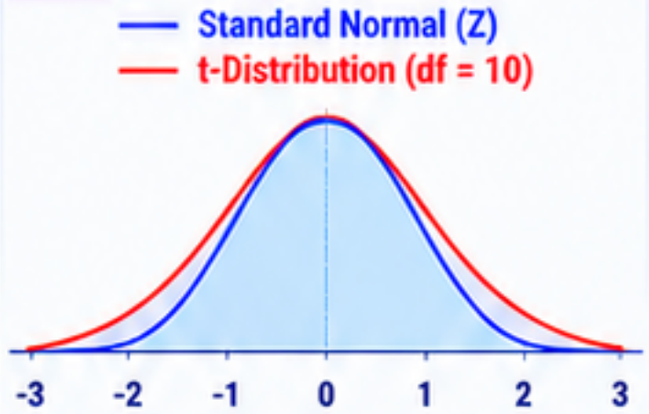
Quick Rule of Thumb

- **Use Z-test** when population standard deviation is known (or sample size is large, $n \geq 30$).
- **Use T-test** when population standard deviation is unknown and sample size is small ($n < 30$).
- Both tests can be used for one-sample, two-sample (independent) and paired comparisons.
- If $n \geq 30$, the t-distribution becomes very similar to the normal distribution, so Z-test can also be used.
- Always check data normality and outliers before selecting the test.

1 Comparison Table: Z-Test vs. T-Test

Aspect	Z-Test	T-Test
Population standard deviation (σ)	Known	Unknown (estimated from sample)
Test statistic	Follows standard normal distribution (Z)	Follows t-distribution (with $n-1$ degrees of freedom)
Sample size (n)	Typically $n \geq 30$ (large sample)	Typically $n < 30$ (small sample)
Distribution shape	Normal distribution	t-distribution (heavier tails)
Degrees of freedom	Not applicable	$n-1$
Use cases	Quality control, large surveys, A/B testing (large sample)	Small sample studies, market research, medical research, before-after analysis

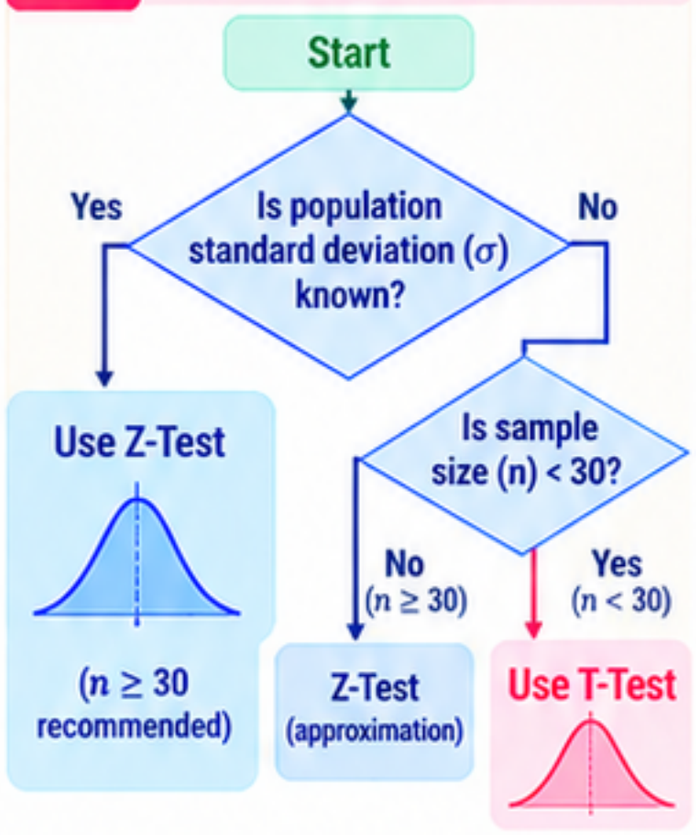
2 Visual Comparison of Distributions



Key Observations

- t-distribution has heavier tails than normal distribution.
- As sample size increases, the t-distribution approaches the normal distribution.
- For large samples ($n \geq 30$), both tests give almost the same results.

3 Decision Flowchart



4 Real-World Examples

Example 1: Z-Test (Population σ Known)

A company produces light bulbs with a known standard deviation of 5 hours. A sample of 50 bulbs has an average lifetime of 48 hours. Is the mean lifetime different from 50 hours?

Use Z-Test (σ known, $n = 50$).

Example 2: T-Test (Population σ Unknown)

A pharmaceutical company tests a new medicine with a sample of 15 patients. The population standard deviation is unknown. We want to check if the mean recovery time is different from 10 days.

Use T-Test (σ unknown, $n = 15$).

Example 3: Large Sample (Z or T)

A retailer correct feedback from 120 customers to check if the average satisfaction score is different from 7. Population standard deviation is unknown, but sample size is large ($n = 120$).

Use Z-Test ($n \geq 30$, $t \approx z$).

5 Summary of When to Use

Scenario	Recommended Test
One sample, σ known	Z-test
One sample, σ unknown, $n < 30$	**T-test**
One sample, σ unknown, $n \geq 30$	Z-test (or T-test, results similar)
Two independent samples, σ known	Z-test
Two independent samples, σ unknown, $n < 30$	**T-test**
Two independent samples, σ unknown, $n \geq 30$	Z-test (or T-test)
Paired samples (before-after)	**T-test**

6 Business Implications

- Using the correct test ensures reliable conclusions and better decision making.
- Helps in quality control, product testing, marketing analysis, and clinical research.
- Avoids false positives (Type I error) and false negatives (Type II error).
- Leads to cost savings, improved processes, and more confident strategic decisions.
- Particularly important for small samples, where the t-test is more appropriate.

7 Key Takeaways

- Check whether σ is known or unknown.
- Consider sample size (n).
- Understand the type of test (one-sample, two-sample, paired).
- Verify assumptions (normality, independence).
- For large samples ($n \geq 30$), Z-test and T-test give similar results.
- Always interpret results in the business context.
- When unsure, T-test is a safer choice for small samples.

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MODULE 9
STATISTICAL TESTS FOR MEANS
From Concepts to Real-World Decisions

"Better Analysts Make a Smarter Tomorrow..."

Z-TEST
T-TEST
REAL DATA
STEP-BY-STEP

11 Excel Implementation of Z-Tests and T-Tests

Step-by-Step Guide with Formulas, Examples, and Business Interpretation

Excel Functions for Z and T Tests

Purpose	Excel Function	Syntax
Z-test (one sample, σ known)	Z.TEST	=Z.TEST(array, x, [σ])
T-test (one sample)	T.TEST or T.DIST.2T	=T.TEST(array, x, tails, type) =T.DIST.2T(|t|, df)
T-test (two sample, independent)	T.TEST	=T.TEST(array1, array2, tails, type)
T-test (two sample, paired)	T.TEST	=T.TEST(array1, array2, tails, type)
Descriptive statistics	AVERAGE, STDEV.S, COUNT	=AVERAGE(range) =STDEV.S(range) =COUNT(range)

2 Data Setup in Excel

Sample Data: Daily Delivery Time (in minutes)

	A	B
1	Observation	Delivery Time
2	1	4.8
3	2	5.1
4	3	5.2
5	4	4.9
5	5	5.3
6	6	5.0
7	7	4.7
8	8	5.1
9	9	4.9
11	10	5.2

3 Descriptive Statistics in Excel

Statistic	Excel Formula	Result
Sample size (n)	=COUNT(B2:B11)	10
Sample mean ($\bar{X}$)	=AVERAGE(B2:B11)	5.02
Sample standard deviation (s)	=STDEV.S(B2:B11)	0.20

Before performing any test, always calculate descriptive statistics to understand your data (mean, standard deviation, sample size).

4 One Sample Z-Test in Excel (σ Known)

Business Problem:
A company claims that the average delivery time is 5 minutes. We want to test if the actual mean is different from 5 minutes. (Population standard deviation $\sigma = 0.5$).

Step	Excel Formula	Result
Sample mean ($\bar{X}$)	=AVERAGE(B2:B11)	5.02
Known σ		0.5
Z-test (two-tailed)	=Z.TEST(B2:B11,5,0.5)	0.7421
Decision ($\alpha = 0.05$)	If p-value > 0.05, Fail to Reject H_0	0.7421 > 0.05 Fail to Reject H_0

Conclusion: There is no significant evidence that the average delivery time is different from 5 minutes.

5 One Sample T-Test in Excel (σ Unknown)

Business Problem:
A call center claims that the average call handling time is 5 minutes. We test if the actual mean is different from 5 minutes.

Step	Excel Formula	Result
Sample mean ($\bar{X}$)	=AVERAGE(B2:B11)	5.02
Sample standard deviation (s)	=STDEV.S(B2:B11)	0.20
T-test (two-tailed)	=T.TEST(B2:B11,5,2,1)	0.7654
Degrees of freedom	=COUNT(B2:B11)-1	9
Decision ($\alpha = 0.05$)	If p-value > 0.05, Fail to Reject H_0	0.7654 > 0.05 Fail to Reject H_0

Conclusion: There is no significant evidence that the average call handling time is different from 5 minutes.

6 Two Sample T-Test (Independent) in Excel

Business Problem:
We want to compare the average sales of two stores (Store A vs Store B) to check if there is a significant difference.

Statistic	Store A (B2:B11)	Store B (D2:D11)
Sample mean ($\bar{X}$)	52.4	49.1
Sample standard deviation (s)	4.8	5.1
Sample size (n)	10	10
Excel Formula:	=T.TEST(B2:B11,D2:D11,2,2)	
Result (p-value):	0.0416	
Decision ($\alpha = 0.05$)	0.0416 < 0.05 Reject H_0	

Conclusion: There is a significant difference between the average sales of Store A and Store B.

7 Paired Sample T-Test in Excel

Business Problem:
We want to check if a training program has improved employee productivity (before vs after).

Statistic	Before (B2:B11)	After (C2:C11)
Sample mean ($\bar{X}$)	68.2	74.5
Sample standard deviation (s)	6.1	5.8
Sample size (n)	10	10
Excel Formula:	=T.TEST(B2:B11,C2:C11,2,1)	
Result (p-value):	0.0127	
Decision ($\alpha = 0.05$)	0.0127 < 0.05 Reject H_0	

Conclusion: The training program has significantly improved employee productivity.

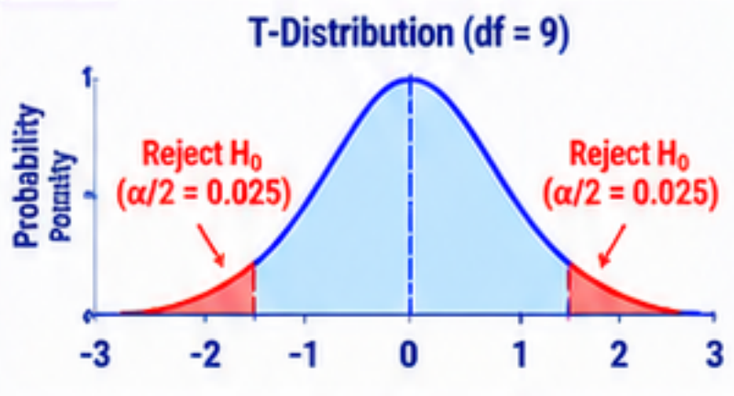
8 Using T.DIST for Manual Calculation

You can also calculate the p-value manually using the t-statistic and degrees of freedom.

Step	Excel Formula	Result
Sample mean ($\bar{X}$)		5.02
Hypothesized mean (μ_0)		5
Sample standard deviation (s)		0.20
Sample size (n)		10
t-statistic	=(5.02-5)/(0.20/SQRT(10))	0.316
Degrees of freedom (df)	10-1	9
p-value (two-tailed)	=T.DIST.2T(ABS(0.316),9)	0.758

T.DIST.2T() gives the two-tailed p-value.
Use T.DIST.RT() for one-tailed tests.

9 Visualizing Results in Excel



Critical Values in Excel

Two-tailed (α = 0.05, df = 9)	=T.INV.2T(0.05,9)
One-tailed (α = 0.05, df = 9)	=T.INV(0.05,9) 2.821

10 Common Excel Tips

- Use Data Analysis Toolpak (T-Test) for quick results.
- Use formula methods for better understanding.
- Always check assumptions (normality, independence).
- Use descriptive statistics before testing.
- Format your data clearly with headers.
- For large datasets, ensure no blank cells or text values.
- Use charts to visualize the distribution and results.

11 Business Use Cases

- Compare average sales between two stores
- Check call handling time against target
- Evaluate production time after process improvement
- Analyze training impact on employee performance
- Compare average revenue before and after a campaign
- Compare patient recovery time between two treatments

12 Key Takeaways

- **Z-test in Excel:** Use Z.TEST() when population standard deviation (σ) is known.
- **T-test in Excel:** Use T.TEST() when σ is unknown.
- Choose the correct test based on data and assumptions.
- Use descriptive statistics to understand your data.
- Interpret the p-value in the business context.
- Excel provides both direct functions and manual methods (T.DIST) for calculations.
- Practice with real business datasets to gain confidence.

12 Practical Applications of Z-Tests and T-Tests

Real-World Business Scenarios and Decision Making

Why These Tests Are Important in Business?

Z-tests and T-tests help businesses make data-driven decisions by checking whether an observed difference, improvement, or change in a metric is statistically significant or just due to random variation.

> "These tests convert business questions into statistical evidence, helping us make confident and objective decisions."

Key Business Questions Answered

1. Has there been a significant improvement after a campaign or intervention?
2. Is the performance of two teams or branches different?
3. Is the average value (sales, cost, time) different from a target?
4. Does a new process reduce defects or delivery time?
5. Is a new product variant performing better than the old one?
6. Are the before-and-after results significantly different?

Which Test to Use?

Scenario	Recommended Test
Compare mean with a target (σ known, large n)	One Sample Z-Test
Compare mean with a target (σ unknown, small n)	One Sample T-Test
Compare means of two independent groups (e.g., two stores)	Independent Two Sample T-Test (or Z-Test if σ known)
Compare means of two related samples (before-after)	Paired Sample T-Test

1 Marketing: Campaign Effectiveness

Problem:
A company wants to check if a new digital marketing campaign has increased the average daily website visits compared to 1,000 visits (before campaign). Population standard deviation is known (σ = 200).

Metric	Value
Sample size (n)	50
Sample mean ($\bar{x}$)	1,120
Population σ	200
Test	One Sample Z-Test

Average Daily Visits

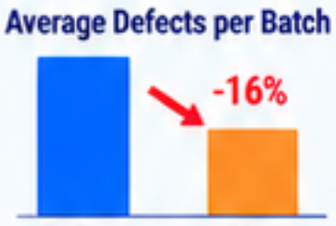
Insight: Since p-value < 0.05, the increase in website visits is statistically significant. The campaign was effective.

2 Operations: Reducing Defects

Problem:
A manufacturing company claims that a new machine has reduced the average number of defects per batch from 5 to less than 5. A sample of 30 batches shows a mean of 4.2 defects with a standard deviation of 1.8 (σ unknown).

Metric	Value
n	30
$\bar{x}$	4.2
s	1.8
Test	One Sample T-Test

Average Defects per Batch



Insight: p-value < 0.05, so the new machine significantly reduces the average number of defects.

3 Sales: Comparing Two Regions

Problem:
A company wants to know if the average monthly sales of Region A and Region B are different. Two independent samples of 25 stores from each region are selected (σ unknown).

Metric	Region A	Region B
n	25	25
$\bar{x}$ (in ₹K)	120	105
s (in ₹K)	18	20
Test	Independent Two Sample T-Test	

Average Monthly Sales (₹ Thousands)

Insight: p-value < 0.05, so there is a significant difference in average monthly sales between the two regions.

4 HR Analytics: Training Effectiveness

Problem:
A company provides a training program to improve employee productivity. We want to check if the average productivity (tasks/week) has increased after training for the same employees.

Metric	Value
n (employees)	15
Before Training ($\bar{x}$)	28
After Training ($\bar{x}$)	34
s (difference)	4.2
Test	Paired Sample T-Test

Average Productivity (Tasks/Week)

Insight: p-value < 0.05, so the training program has significantly improved employee productivity.

5 Customer Experience: Service Time

Problem:
A call center wants to check if the average call handling time is now less than 5 minutes after process improvement. A sample of 20 calls shows a mean of 4.3 minutes with a standard deviation of 1.1 (σ unknown).

Metric	Value
n	20
$\bar{x}$ (minutes)	4.3
s (minutes)	1.1
Test	One Sample T-Test

Average Call Handling Time (minutes)

Insight: p-value < 0.05, so the average call handling time is significantly lower than 5 minutes.

6 Product Analytics: A/B Testing

Problem:
An e-commerce company tests two versions of a product page (A and B) to check if the average conversion rate is different. Two independent samples are taken (σ unknown).

Metric	Version A	Version B
n	30	30
$\bar{x}$ (%)	2.8	3.4
s (%)	0.9	1.1
Test	Independent Two Sample T-Test	

Conversion Rate (%)

Insight: p-value < 0.05, so Version B has a significantly higher average conversion rate than Version A.

More Business Use Cases

Finance
Compare average loan amounts between customer segments.

Retail
Compare average purchase value between online and offline stores.

Healthcare
Compare average recovery time between two treatments.

Education
Compare test scores before and after a new teaching method.

Supply Chain
Compare average delivery time between two logistics partners.

Quality Control
Check if the average product weight meets the target specification.

Key Takeaways

- Z-tests and T-tests help validate business claims with statistical evidence.
- Choose the right test based on data type, sample size, and whether σ is known.
- A p-value < 0.05 indicates a statistically significant result (at 5% significance level).
- Always check assumptions before performing the test.
- Interpret the results in business context, not just statistically.
- These tests are widely used in marketing, sales, operations, HR, finance, and product analytics.

Common Mistakes to Avoid

- Using the wrong test for the scenario.
- Ignoring assumptions (normality, independence).
- Misinterpreting p-value (p < 0.05 does not mean the effect is large).
- Using small sample Z-test (should use T-test instead).
- Not checking for outliers.
- Confusing statistical significance with business significance.
- Not considering practical impact (e.g., cost, ROI).

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MODULE 9

STATISTICAL TESTS FOR MEANS

From Concepts to Real-World Decisions

"Better Analysts Make a Smarter Tomorrow..."

FINAL SUMMARY
Key Concepts | Formulas
Business Applications
Interview Questions

Key Takeaways from this Module

1. Z-test and T-test are used to test hypotheess about population means.
2. Z-test is used when population standard deviation (σ) is known and sample size is large ($n \geq 30$).
3. T-test is used when population standard deviation is unknown and sample size is small ($n < 30$).
4. T-tests include one-sample, two-sample (independent) and paired sample tests.
5. The t-distribution has heavier tails than the normal distribution, especially for small sample sizes.
6. Assumptions (normality, independence, no extreme outliers) are crucial for valid results.
7. p-value helps us decide whether to reject or fail to reject the null hypothesis.
8. These tests have wide business applications in marketing, sales, operations, HR, finance, healthcare and product analytics.

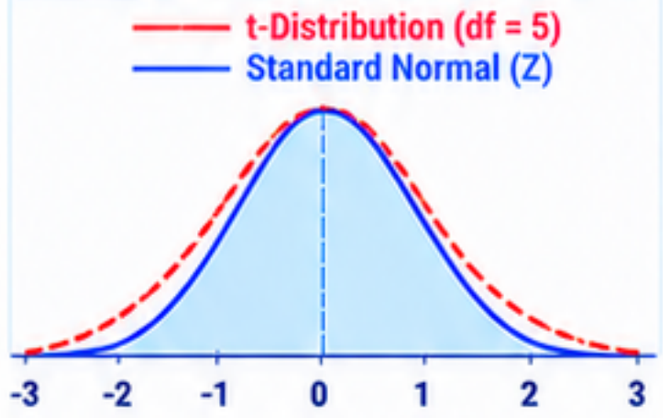
Key Formulas Summary

Test Type	Test Statistic Formula	Degrees of Freedom (df)
Z-test (One Sample)	$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	Not applicable
T-test (One Sample)	$t = \frac{\bar{X} - \mu}{s / \sqrt{n}}$	$df = n - 1$
T-test (Two Sample Independent)	$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	$df = n_1 + n_2 - 2$
T-test (Paired Sample)	$t = \frac{\bar{d}}{s_d / \sqrt{n}}$	$df = n - 1$

Z-Test vs. T-Test Quick Comparison

Aspect	Z-Test	T-Test
Population standard deviation (σ)	Known	Unknown (estimated from sample)
Sample size (n)	Typically $n \geq 30$ (large sample)	Typically $n < 30$ (small sample)
Test statistic	Follows standard normal distribution (Z)	Follows t-distribution (with $n - 1$ df)
Distribution shape	Normal distribution	t-distribution (heavier tails)
Degrees of freedom	Not applicable	$n - 1$
Common use cases	Large surveys, A/B testing, quality control	Small sample studies, market research, before-after analysis

Distribution Comparison



The t-distribution has heavier tails than the normal distribution, which accounts for extra uncertainty when the sample size is small.

Assumptions of T-Tests

- Population standard deviation is unknown.
- Sample is randomly selected.
- Observations are independent (or correctly paired for paired t-test).
- Data is approximately normally distributed (or sample size is large: $n \geq 30$).
- No extreme outliers in the data.
- For two-sample t-test: the two samples are independent (unpaired) or correctly paired.

Business Applications Across Industries

Marketing
- **Campaign** effectiveness
- Compare conversion rates (A/B testing)
- Check lift in average revenue per user

Sales
- **Compare** sales across regions/stores
- Evaluate impact of new sales strategy
- Check improvement in average deal size

Operations
- **Reduce** defects in manufacturing
- Improve process efficiency
- Compare average processing time

HR Analytics
- **Training** effectiveness (before vs after)
- Compare employee performance
- Evaluate impact on produstivity and attrition

Finance
- **Compare** average loan amounts
- **Evaluate** cost reduction initiatives
- Analyze change in average revenue or profit margin

Healthcare
- **Compare** patient recovery time
- Evaluate treatment effectiveness
- Check improvement in average wait time

E-commerce
- **Compare** average order value
- Test new product variant performance
- Analyze impact on customer satisfaction

Common Mistakes to Avoid

- Using the wrong test (Z vs T).
- Ignoring assumptions (normality, independence).
- Not checking for outliers.
- Misinterpreting p-value ($p < 0.05$ does not mean the effect is large).
- Using small sample Z-test (should use T-test).
- Not considering **business** impact (statistical significance vs. practical significance).
- Using one-tailed test when two-tailed is appropriate.

Result Interpretation

- If p-value < α (e.g., 0.05): Reject H_0 → Statistically significant difference exists.
- If p-value ≥ α: Fail to reject H_0 → No statistically significant difference.
- Look at **mean difference** to understand direction and magnitude.
- Consider business context, not just statistical significance.
- Check confidence intervals for additional insight (if covered).

Interview Questions with Answers

Q1 When should you use a Z-test instead of a T-test?
A Use a Z-test when the population standard deviation (σ) is known and the sample size is large ($n \geq 30$).

Q2 What is the main reason for using a T-test?
A A T-test is used when the population standard deviation is unknown and the sample size is small ($n < 30$).

Q3 What are the assumptions of a T-test?
A Random sampling, independence, approximate normality (or large n), and no extreme outliers.

Q4 How do you interpret a p-value in hypothesis testing?
A If p-value < 0.05, we reject the null hypothesis (H_0). It means there is statistically significant evidence against H_0.

Q5 Which test would you use to compare before and after results on the same group?
A Use a paired sample T-test, as the observations are related.

Final Thoughts

"Statistical tests are not just formulas — they are tools to convert data into evidence and evidence into better business decisions."